

SAT Math Mastery · Week 4B

Polynomial, Rational & Radical Equations

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics factor theorem and remainders · polynomial structure and zeros · rational equations · radical equations and extraneous solutions · equivalent expressions and exponents · literal equations · building expressions

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Use the factor theorem and remainder theorem: $(x - r)$ is a factor $\iff p(r) = 0$; remainder on division by $x - r$ is $p(r)$.
- S2.** Read zeros and factors from tables and graphs; match a graph (crossing, bouncing, end behaviour) to an equation.
- S3.** Solve rational equations by clearing denominators and discard solutions that make a denominator 0.
- S4.** Solve radical equations by isolating the radical and squaring; check every candidate in the original equation.
- S5.** Rewrite and combine rational expressions; simplify expressions with rational exponents and radicals.
- S6.** Rearrange literal equations and formulas to isolate a chosen variable.
- S7.** Use structure: substitution $u = x^2$, symmetric expressions, Vieta for cubics.
- S8.** Build one-variable expressions from geometry or word contexts (area, volume, rates) and solve.

2 · REQUIRED FACTS

- Factor theorem: $(x - r)$ is a factor of $p(x) \iff p(r) = 0$. Remainder theorem: $p(x) \div (x - r)$ leaves remainder $p(r)$.
- A table entry $p(r) = 0$ gives the factor $(x - r)$; a zero that *touches* the axis without crossing has even multiplicity.
- A rational expression is undefined where its denominator is 0; these values are excluded from every solution set.
- Squaring both sides of an equation can create *extraneous* solutions. Always substitute back, and remember $\sqrt{} \geq 0$.
- $a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$, $a^{-n} = \frac{1}{a^n}$, $a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$.
- To isolate a variable, undo operations in reverse order; if the target appears twice, gather those terms and factor it out.

3 · REQUIRED FORMULAS

$$p(x) = (x - r)q(x) + p(r), \quad (x - r) \mid p(x) \iff p(r) = 0$$

$$x^3 + bx^2 + cx + d: \quad r_1 + r_2 + r_3 = -b, \quad r_1 r_2 + r_1 r_3 + r_2 r_3 = c, \quad r_1 r_2 r_3 = -d$$

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}, \quad \sqrt{x} = x - k \Rightarrow x = (x - k)^2 \text{ (then check } x - k \geq 0), \quad r^2 + s^2 + t^2 = (r + s + t)^2 - 2(rs + rt + st)$$

4 · MINI THEORY SHEET

Factor and remainder. To test a candidate factor $(x - r)$, evaluate $p(r)$; if it is 0 the factor works. If the remainder is given, set $p(r) = \text{remainder}$ and solve for the constant.

Rational equations. (1) Note the excluded values. (2) Multiply every term by the common denominator. (3) Solve. (4) Reject any solution equal to an excluded value. If every solution is rejected, there is *no solution*.

Radical equations. (1) Isolate one radical. (2) Square both sides (expand $(x - k)^2$ fully). (3) Solve. (4) Check each candidate in the *original* equation: the radical side must be ≥ 0 .

Equivalent expressions. Put fractions over a common denominator; write radicals as fractional exponents, combine, then convert back. Test a choice by plugging in a convenient number.

Literal equations. Treat other letters as numbers. Clear fractions first, collect the target variable on one side, factor, divide. Roots and squares are undone by squaring and square-rooting.

Graph to equation. Zeros give factors. A graph that bounces at a zero has a squared factor. End behaviour fixes the sign of the leading coefficient. A vertical asymptote at $x = c$ means $(x - c)$ is in the denominator; the horizontal asymptote is the ratio of leading coefficients.

Speed shortcuts.

Situation	Shortcut
“is $(x - r)$ a factor?”	compute $p(r)$ only: no division
remainder on division by $(x - r)(x - s)$	remainder is $ax + b$; use $p(r)$ and $p(s)$ as two equations
zeros of a monic cubic, constant d	product of zeros $= -d$; sum $= -b$
$x^4 + bx^2 + c$	substitute $u = x^2$; real x needs $u \geq 0$
$\sqrt{x} + \frac{1}{\sqrt{x}} = s$	square: $x + \frac{1}{x} = s^2 - 2$
radical equation answer choices	plug in the choices; the extraneous one fails
rational equation with a parameter	solve for x , then ask when x hits the excluded value
rewriting with exponents	test with $x = 2$ or $x = 4$ on both sides

5 · GUIDED EXAMPLES

Example 1 — Factor theorem to find a constant

$x + 3$ is a factor of $p(x) = x^3 + kx^2 - 4x + 3$. Find k .

$x + 3 = x - (-3)$, so $p(-3) = 0$: $-27 + 9k + 12 + 3 = 0 \Rightarrow 9k = 12 \Rightarrow k = \frac{4}{3}$.

Example 2 — Rational equation and an excluded value

Solve $\frac{x^2}{x - 4} = \frac{16}{x - 4}$.

Excluded value: $x = 4$. Multiply by $(x - 4)$: $x^2 = 16 \Rightarrow x = \pm 4$. The solution $x = 4$ makes the denominator 0, so reject it. The only solution is $x = -4$.

Example 3 — Radical equation and an extraneous solution

Solve $\sqrt{x + 6} = x$.

Square: $x + 6 = x^2 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0$. Check $x = 3$: $\sqrt{9} = 3$ ✓. Check $x = -2$: $\sqrt{4} = 2 \neq -2$, extraneous. The only solution is $x = 3$.

6 · PRACTICE SET — 48 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + x^2 - 10x + 8$ has exactly one of the following as a factor. Which one?

(A) $x + 2$

(B) $x - 1$ ✓

(C) $x + 1$

(D) $x - 4$

1. A binomial $(x - r)$ is a factor exactly when $p(r) = 0$.

2. $p(1) = 1 + 1 - 10 + 8 = 0$, so $x - 1$ is a factor.

3. The others fail: $p(-1) = 1 - 1 + 10 + 8 = 18$, $p(4) = 64 + 16 - 40 + 8 = 48$, $p(-2) = -8 + 4 + 20 + 8 = 24$.

Answer: (B)

2. EASY · NO CALCULATOR NEEDED

When $f(x) = x^3 - 3x + 5$ is divided by $x - 2$, what is the remainder?

1. By the remainder theorem, the remainder is $f(2)$.

2. $f(2) = 8 - 6 + 5 = 7$.

Answer: 7

3. EASY · NO CALCULATOR NEEDED

Which of the following expressions is equivalent to $\frac{2x^2 - 18}{x - 3}$ for all $x \neq 3$?

(A) $2x + 3$

(B) $x + 3$

(C) $2x + 6$ ✓

(D) $2x - 6$

1. Factor the numerator: $2x^2 - 18 = 2(x^2 - 9) = 2(x - 3)(x + 3)$.

2. Cancel $(x - 3)$: $2(x + 3) = 2x + 6$.

Answer: (C)

4. EASY · NO CALCULATOR NEEDED

What is the solution to the equation $\sqrt{3x + 4} = 7$?

1. Square both sides: $3x + 4 = 49$.

2. $3x = 45 \Rightarrow x = 15$. (Check: $\sqrt{49} = 7$.)

Answer: 15

5.

EASY · NO CALCULATOR NEEDED

The volume of a cone is $V = \frac{1}{3}\pi r^2 h$. Which of the following gives h in terms of V and r ?

(A) $h = \frac{3V}{\pi r^2}$ ✓

(B) $h = \frac{V}{3\pi r^2}$

(C) $h = 3V\pi r^2$

(D) $h = \frac{3V}{\pi r}$

1. Multiply both sides by 3: $3V = \pi r^2 h$.

2. Divide both sides by πr^2 : $h = \frac{3V}{\pi r^2}$.

Answer: (A)

6.

MEDIUM · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + kx^2 - 5x - 6$, where k is a constant, has $x - 2$ as a factor. What is the value of k ?

1. $x - 2$ a factor means $p(2) = 0$.

2. $8 + 4k - 10 - 6 = 0 \Rightarrow 4k = 8 \Rightarrow k = 2$.

Answer: 2

7.

MEDIUM · NO CALCULATOR NEEDED

The table gives selected values of a polynomial function p .

x	-3	-1	2	4
$p(x)$	5	0	9	-2

Which of the following must be a factor of $p(x)$?

(A) $x + 3$

(B) $x - 4$

(C) $x - 1$

(D) $x + 1$ ✓

1. $(x - r)$ is a factor exactly when $p(r) = 0$.

2. The table shows $p(-1) = 0$, so $x - (-1) = x + 1$ is a factor.

3. The other options need $p(1)$, $p(-3)$ or $p(4)$ to be 0; the table gives $p(-3) = 5$ and $p(4) = -2$, and $p(1)$ is not known to be 0.

Answer: (D)

8.

MEDIUM · NO CALCULATOR NEEDED

What is the solution set of the equation $\frac{x}{x-4} = \frac{4}{x-4} + 3$?

(A) $\{-4\}$ (B) $\{2\}$ **(C) No solution ✓**(D) $\{4\}$ 1. Excluded value: $x = 4$.2. Multiply by $(x - 4)$: $x = 4 + 3(x - 4) \Rightarrow x = 3x - 8 \Rightarrow x = 4$.3. But $x = 4$ makes the denominators 0, so it is extraneous. There is no solution.**Answer: (C)**

9.

MEDIUM · NO CALCULATOR NEEDED

What is the solution to $\sqrt{x+7} = x-5$?

1. Square: $x+7 = x^2 - 10x + 25 \Rightarrow x^2 - 11x + 18 = 0 \Rightarrow (x-2)(x-9) = 0$.2. Check $x = 2$: $\sqrt{9} = 3$ but $x - 5 = -3$. Extraneous.3. Check $x = 9$: $\sqrt{16} = 4 = 9 - 5$. ✓ The solution is 9.**Answer: 9**

10.

MEDIUM · NO CALCULATOR NEEDED

Which of the following is equivalent to $\frac{\sqrt[3]{x^5}}{x}$, for $x > 0$?

(A) $x^{3/2}$ **(B) $x^{2/3}$ ✓**(C) $x^{5/3}$ (D) $x^{-2/3}$ 1. $\sqrt[3]{x^5} = x^{5/3}$.2. $\frac{x^{5/3}}{x^1} = x^{5/3-1} = x^{2/3}$.**Answer: (B)**

11.

MEDIUM · NO CALCULATOR NEEDED

The period of a pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$, where L is its length and g is the acceleration due to gravity. Which of the following gives L in terms of T and g ?

(A) $L = \frac{gT^2}{2\pi}$

(B) $L = \frac{gT}{2\pi}$

(C) $L = \frac{4\pi^2}{gT^2}$

(D) $L = \frac{gT^2}{4\pi^2}$ ✓

1. Divide by 2π : $\frac{T}{2\pi} = \sqrt{\frac{L}{g}}$.

2. Square both sides: $\frac{T^2}{4\pi^2} = \frac{L}{g}$.

3. Multiply by g : $L = \frac{gT^2}{4\pi^2}$.

Answer: (D)

12.

MEDIUM · NO CALCULATOR NEEDED

How many distinct real solutions does the equation $x^4 - 5x^2 - 36 = 0$ have?

(A) 2 ✓

(B) 3

(C) 4

(D) 0

1. Let $u = x^2$: $u^2 - 5u - 36 = 0 \Rightarrow (u - 9)(u + 4) = 0$.

2. $u = 9 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$. $u = -4 \Rightarrow x^2 = -4$ has no real solution.

3. There are 2 distinct real solutions.

Answer: (A)

13.

MEDIUM · NO CALCULATOR NEEDED

A closed rectangular box has a square base. Its height is 2 centimeters more than the side length of its base, and its volume is 96 cubic centimeters. What is the total surface area, in square centimeters, of the box?

1. Let the base side be s ; the height is $s + 2$. Volume: $s^2(s + 2) = 96$.

2. Try $s = 4$: $16 \cdot 6 = 96$. ✓ ($s^3 + 2s^2$ is increasing for $s > 0$, so $s = 4$ is the only positive solution.)

3. Surface area = $2s^2 + 4s(s + 2) = 2(16) + 4(4)(6) = 32 + 96 = 128$.

Answer: 128

14.

HARD · NO CALCULATOR NEEDED

When the polynomial $p(x)$ is divided by $x - 3$, the remainder is 5. When $p(x)$ is divided by $x + 1$, the remainder is -3 . When $p(x)$ is divided by $(x - 3)(x + 1)$, the remainder is $ax + b$, where a and b are constants. What is the value of $a + b$?

1. Write $p(x) = (x - 3)(x + 1)q(x) + ax + b$.
2. Remainder theorem: $p(3) = 5 \Rightarrow 3a + b = 5$ and $p(-1) = -3 \Rightarrow -a + b = -3$.
3. Subtract: $4a = 8 \Rightarrow a = 2$, then $b = -1$.
4. $a + b = 2 + (-1) = 1$.

Answer: 1

15.

HARD · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + ax^2 + bx - 12$, where a and b are constants, has zeros at $x = 1$ and $x = -3$. What is the third zero of p ?

(A) -12 **(B) -4 ✓**(C) 4 (D) 12

1. For a monic cubic, the product of the zeros is $-d = -(-12) = 12$.
2. $1 \cdot (-3) \cdot r = 12 \Rightarrow r = -4$.
3. (Check: $p(1) = 0$ and $p(-3) = 0$ give $a = 6$, $b = 5$; then $p(-4) = -64 + 96 - 20 - 12 = 0$.)

Answer: (B)

16.

HARD · NO CALCULATOR NEEDED

What is the solution to the equation $\frac{x}{x-3} + \frac{2}{x+1} = \frac{12}{x^2 - 2x - 3}$?

1. $x^2 - 2x - 3 = (x - 3)(x + 1)$. Excluded values: $x = 3$ and $x = -1$.
2. Multiply by $(x - 3)(x + 1)$: $x(x + 1) + 2(x - 3) = 12 \Rightarrow x^2 + 3x - 18 = 0 \Rightarrow (x + 6)(x - 3) = 0$.
3. $x = 3$ is excluded, so the only solution is $x = -6$. (Check: $\frac{-6}{-9} + \frac{2}{-5} = \frac{4}{15}$ and $\frac{12}{45} = \frac{4}{15}$.)

Answer: -6

17.

HARD · NO CALCULATOR NEEDED

What is the solution to the equation $\sqrt{x+4} - \sqrt{x-1} = 1$?

1. Isolate one radical: $\sqrt{x+4} = 1 + \sqrt{x-1}$.
2. Square: $x+4 = 1 + 2\sqrt{x-1} + (x-1) \Rightarrow 4 = 2\sqrt{x-1}$.
3. $\sqrt{x-1} = 2 \Rightarrow x-1 = 4 \Rightarrow x = 5$. (Check: $\sqrt{9} - \sqrt{4} = 3 - 2 = 1$.)

Answer: 5**18.**

HARD · NO CALCULATOR NEEDED

Which of the following is equivalent to $\frac{2}{x-1} - \frac{3}{x+2}$?

(A) $\frac{-1}{(x-1)(x+2)}$

(B) $\frac{7-x}{2x+1}$

(C) $\frac{7-x}{(x-1)(x+2)}$ ✓

(D) $\frac{1-x}{(x-1)(x+2)}$

1. Common denominator $(x-1)(x+2)$: $\frac{2(x+2) - 3(x-1)}{(x-1)(x+2)}$.

2. Numerator: $2x+4-3x+3 = 7-x$.

3. Result: $\frac{7-x}{(x-1)(x+2)}$.

Answer: (C)

19.

HARD · NO CALCULATOR NEEDED

The formula $P = \frac{4Q}{R+Q}$ relates the positive quantities P , Q , and R , where $P < 4$. Which of the following gives Q in terms of P and R ?

(A) $Q = \frac{PR}{4-P}$ ✓

(B) $Q = \frac{PR}{P-4}$

(C) $Q = \frac{PR}{P+4}$

(D) $Q = \frac{(4-P)R}{P}$

1. Clear the denominator: $P(R+Q) = 4Q \Rightarrow PR + PQ = 4Q$.

2. Collect Q -terms: $PR = 4Q - PQ = Q(4-P)$.

3. $Q = \frac{PR}{4-P}$.

Answer: (A)

20.

HARD · NO CALCULATOR NEEDED

The polynomial $x^3 - 7x^2 + 14x - 8$ has three real zeros r , s , and t . What is the value of $r^2 + s^2 + t^2$?

1. Vieta for a cubic: $r + s + t = 7$ and $rs + rt + st = 14$.

2. $r^2 + s^2 + t^2 = (r + s + t)^2 - 2(rs + rt + st) = 49 - 28 = 21$.

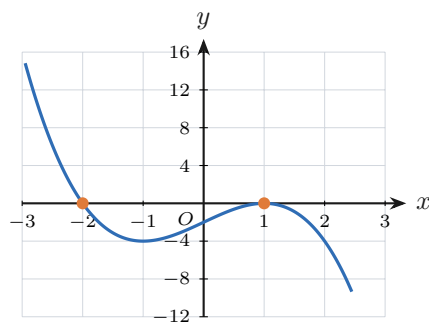
3. (Check: the zeros are 1, 2, 4 and $1 + 4 + 16 = 21$.)

Answer: 21

21.

HARD · DESMOS

The graph of the polynomial function p is shown in the xy -plane. Which of the following could define $p(x)$?



(A) $p(x) = (x + 2)(x - 1)^2$

(B) $p(x) = -(x - 2)(x + 1)^2$

(C) $p(x) = -(x + 2)(x - 1)$

(D) $p(x) = -(x + 2)(x - 1)^2$ ✓

1. The graph crosses the x -axis at $x = -2$ (single factor $x + 2$) and touches it at $x = 1$ (squared factor $(x - 1)^2$).
2. For large positive x the graph goes down and for large negative x it goes up, so the leading coefficient is negative:
 $p(x) = -(x + 2)(x - 1)^2$.
3. Check the y -intercept: $p(0) = -(2)(1) = -2$, matching the graph.

Answer: (D)

22.

HARD · NO CALCULATOR NEEDED

The length of a rectangle is 3 inches more than its width. The diagonal of the rectangle is 15 inches. What is the area of the rectangle, in square inches?

(A) 90

(B) 96

(C) 108 ✓

(D) 120

1. Let the width be w ; the length is $w + 3$. Pythagorean theorem: $w^2 + (w + 3)^2 = 15^2$.
2. $2w^2 + 6w + 9 = 225 \Rightarrow w^2 + 3w - 108 = 0 \Rightarrow (w + 12)(w - 9) = 0$. Width is positive, so $w = 9$.
3. The length is 12, and the area is $9 \cdot 12 = 108$.

Answer: (C)

23.

HARD · NO CALCULATOR NEEDED

For $x > 0$, which of the following is equivalent to $\frac{(27x^9)^{2/3}}{x^{-2}\sqrt{81x^4}}$?

(A) x^{10}

(B) x^6 ✓

(C) $2x^6$

(D) $9x^6$

1. Numerator: $(27x^9)^{2/3} = 27^{2/3} x^6 = 9x^6$.
2. Denominator: $\sqrt{81x^4} = 9x^2$, so $x^{-2} \cdot 9x^2 = 9$.
3. $\frac{9x^6}{9} = x^6$.

Answer: (B)**24.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + ax^2 + bx + c$, where a , b , and c are constants, has $x - 1$ and $x + 2$ as factors. When $p(x)$ is divided by $x - 3$, the remainder is 40. What is the value of $p(4)$?

1. p is monic cubic with two known factors, so $p(x) = (x - 1)(x + 2)(x - r)$ for some number r .
2. Remainder theorem: $p(3) = 40 \Rightarrow (2)(5)(3 - r) = 40 \Rightarrow 3 - r = 4 \Rightarrow r = -1$.
3. So $p(x) = (x - 1)(x + 2)(x + 1)$.
4. $p(4) = (3)(6)(5) = 90$.

Answer: 90

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

For which of the following values of k does the equation $\frac{2x}{x-4} - \frac{k}{x-4} = 3$ have no solution?

(A) 4

(B) 12

(C) -8

(D) 8 ✓

1. Excluded value: $x = 4$. Multiply by $(x - 4)$: $2x - k = 3(x - 4) = 3x - 12$.
2. Solve: $x = 12 - k$. This is the only candidate.
3. There is no solution exactly when the candidate is the excluded value: $12 - k = 4 \Rightarrow k = 8$.
4. (For $k = 4$ the solution is $x = 8$; for $k = 12$ it is $x = 0$; for $k = -8$ it is $x = 20$.)

Answer: (D)**26.**

SAT CHALLENGE · NO CALCULATOR NEEDED

If $x > 0$ and $\sqrt{x} + \frac{1}{\sqrt{x}} = 5$, what is the value of $x + \frac{1}{x}$?

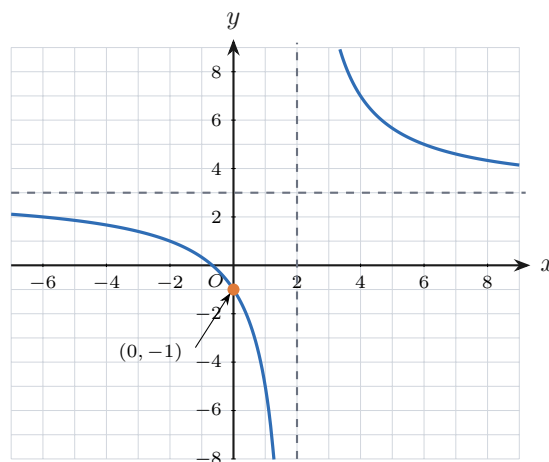
1. Square both sides: $\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = 25$.
2. Expand: $x + 2\sqrt{x} \cdot \frac{1}{\sqrt{x}} + \frac{1}{x} = x + 2 + \frac{1}{x} = 25$.
3. $x + \frac{1}{x} = 23$.

Answer: 23

27.

SAT CHALLENGE · DESMOS

The graph of $f(x) = \frac{ax+b}{x-c}$ is shown in the xy -plane, where a , b , and c are constants. The dashed lines are the asymptotes of the graph, and the graph passes through $(0, -1)$. What is the value of $a + b + c$?



1. The vertical asymptote is $x = 2$; the factor $x - c$ vanishes there, so $c = 2$.
2. The horizontal asymptote is $y = 3$, the ratio of leading coefficients, so $a = 3$.
3. $f(0) = \frac{b}{0-2} = -1 \Rightarrow b = 2$.
4. $a + b + c = 3 + 2 + 2 = 7$. (Check: $f(4) = \frac{14}{2} = 7$ and $f(1) = \frac{5}{-1} = -5$ lie on the plotted curve.)

Answer: 7

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

For which of the following positive values of k does the equation $x^4 - 10x^2 + k = 0$ have exactly two distinct real solutions?

(A) 25 ✓

(B) 50

(C) 5

(D) 10

- Let $u = x^2$: $u^2 - 10u + k = 0$. Each positive solution u gives two real values $x = \pm\sqrt{u}$; $u < 0$ gives none.
- Exactly two real x requires exactly one positive u : the repeated solution, $\Delta = 100 - 4k = 0 \Rightarrow k = 25$ ($u = 5$, $x = \pm\sqrt{5}$). (For $k > 0$ both u -solutions have the same sign, so two distinct $u > 0$ would give four real x .)
- Check the others: $k = 5$ and $k = 10$ give two positive u (four real x); $k = 50$ gives $\Delta < 0$ (none).

Answer: (A)

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

The positive quantities u , a , and b satisfy $\frac{1}{\sqrt{u}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$. Which of the following gives u in terms of a and b ?

(A) $u = \frac{(\sqrt{a} + \sqrt{b})^2}{ab}$

(B) $u = \frac{ab}{(\sqrt{a} + \sqrt{b})^2}$ ✓

(C) $u = \frac{ab}{a + b}$

(D) $u = \frac{\sqrt{ab}}{\sqrt{a} + \sqrt{b}}$

- Add the fractions: $\frac{1}{\sqrt{u}} = \frac{\sqrt{b} + \sqrt{a}}{\sqrt{a}\sqrt{b}}$.
- Take reciprocals: $\sqrt{u} = \frac{\sqrt{ab}}{\sqrt{a} + \sqrt{b}}$.
- Square both sides: $u = \frac{ab}{(\sqrt{a} + \sqrt{b})^2}$.

Answer: (B)

30.

SAT CHALLENGE · CALCULATOR

A boat travels 24 kilometers upstream and then 36 kilometers downstream in a river whose current flows at 3 kilometers per hour. The entire trip takes 7 hours. What is the speed of the boat in still water, in kilometers per hour?

1. Let the boat's speed in still water be $v > 3$. Upstream speed $v - 3$, downstream speed $v + 3$.

2. Time: $\frac{24}{v-3} + \frac{36}{v+3} = 7$. Multiply by $(v-3)(v+3)$: $24(v+3) + 36(v-3) = 7(v^2-9)$.

3. $60v - 36 = 7v^2 - 63 \Rightarrow 7v^2 - 60v - 27 = 0 \Rightarrow (7v+3)(v-9) = 0$.

4. $v = -\frac{3}{7}$ is rejected (the speed must exceed 3), so $v = 9$. (Check: $\frac{24}{6} + \frac{36}{12} = 4 + 3 = 7$.)

Answer: 9

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

For constants p , q and c , the equation

$$(2x-3)(x^2+px+q) = 2x^3 - 5x^2 + cx + 9$$

is true for all real values of x . What is the value of c ?

1. The left side is 0 when $x = \frac{3}{2}$, so the right side must be 0 there too.

2. $2\left(\frac{27}{8}\right) - 5\left(\frac{9}{4}\right) + c\left(\frac{3}{2}\right) + 9 = 0$.

3. $\frac{27}{4} - \frac{45}{4} + 9 + \frac{3}{2}c = 0 \Rightarrow \frac{9}{2} + \frac{3}{2}c = 0 \Rightarrow c = -3$.

4. (Check: $p = -1$, $q = -3$ gives $(2x-3)(x^2-x-3) = 2x^3 - 5x^2 - 3x + 9$.)

Answer: -3

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a positive real number x , $\frac{\sqrt[3]{x} \cdot \sqrt[4]{x^3}}{\sqrt[6]{x}} = \sqrt[3]{k}$, where $k = 2^{11}$. Which of the following is equal to x ?

(A) $2^{\frac{121}{36}}$

(B) $2^{\frac{44}{3}}$

(C) 2^4 ✓

(D) $2^{\frac{11}{3}}$

1. Rational exponents: $x^{1/3} \cdot x^{3/4} \cdot x^{-1/6}$.

2. Add: $\frac{1}{3} + \frac{3}{4} - \frac{1}{6} = \frac{4+9-2}{12} = \frac{11}{12}$, so the left side is $x^{11/12}$.

3. Right side: $\sqrt[3]{2^{11}} = 2^{11/3}$.

4. $x^{11/12} = 2^{11/3}$. Raise both sides to the reciprocal power $\frac{12}{11}$: $x = 2^{\frac{11}{3} \cdot \frac{12}{11}} = 2^4$.

Answer: (C)**33.**

SAT CHALLENGE · NO CALCULATOR NEEDED

For $x > 0$ and any positive integer n , which of the following is equivalent to $\frac{(x^{2n+1})^3 + 4x^{6n+3} - 2x(x^{3n+1})^2}{x^{6n+1}}$?

(A) $3x^2$ ✓

(B) $5x^2$

(C) $3x^4$

(D) $3x^{12n+4}$

1. Power of a power: $(x^{2n+1})^3 = x^{6n+3}$ and $x(x^{3n+1})^2 = x \cdot x^{6n+2} = x^{6n+3}$.

2. The numerator is $x^{6n+3}(1 + 4 - 2) = 3x^{6n+3}$.

3. Divide, subtracting exponents: $3x^{(6n+3)-(6n+1)} = 3x^2$.

Answer: (A)

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

For all $x > 1$,

$$\frac{a}{x-1} - \frac{bx+c}{x^2+x+1} = \frac{3x^2+8x+4}{x^3-1}$$

where a , b , and c are constants. What is the value of c ?

1. Since $x^3 - 1 = (x - 1)(x^2 + x + 1)$, multiply both sides by $x^3 - 1$: $a(x^2 + x + 1) - (bx + c)(x - 1) = 3x^2 + 8x + 4$.
2. Expand the subtracted product with its whole sign: $(bx + c)(x - 1) = bx^2 + (c - b)x - c$. So the left side is $(a - b)x^2 + (a - c + b)x + (a + c)$.
3. Match: x^2 : $a - b = 3$; x : $a + b - c = 8$; constant: $a + c = 4$.
4. From the constant equation $c = 4 - a$; from the first $b = a - 3$. Then $a + (a - 3) - (4 - a) = 8 \Rightarrow 3a = 15$, $a = 5$, so $b = 2$ and $c = -1$.

Answer: -1 **35.**

SAT CHALLENGE · NO CALCULATOR NEEDED

For $y \neq 0$ and $y \neq \frac{1}{3}$, which of the following is equivalent to

$$\frac{\frac{1}{y^3} - 27}{\frac{1}{y} - 3} ?$$

(A) $\frac{1}{y^2} - \frac{3}{y} + 9$

(B) $\frac{1}{y^2} + 9$

(C) $\frac{1}{y^2} + \frac{6}{y} + 9$

(D) $\frac{1}{y^2} + \frac{3}{y} + 9$ ✓

1. Let $u = \frac{1}{y}$. The expression is $\frac{u^3 - 27}{u - 3} = \frac{u^3 - 3^3}{u - 3}$.
2. Difference of cubes: $u^3 - 27 = (u - 3)(u^2 + 3u + 9)$. The $u - 3$ cancels (allowed because $y \neq \frac{1}{3}$).
3. Result: $u^2 + 3u + 9 = \frac{1}{y^2} + \frac{3}{y} + 9$.
4. Check with $y = 1$: $\frac{1 - 27}{1 - 3} = 13$ and $1 + 3 + 9 = 13$. Check with $y = -1$: $\frac{-1 - 27}{-1 - 3} = 7$ and $1 - 3 + 9 = 7$.

Answer: (D)

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 - 9x^2 + kx - 12$ has a positive integer constant k . For some positive integer n , $x - n$ is a factor of $p(x)$. What is the sum of all possible values of k ?

1. If $x - n$ is a factor, then $p(n) = 0$: $n^3 - 9n^2 + kn - 12 = 0$, so $kn = 12 + 9n^2 - n^3$ and $k = \frac{12}{n} + 9n - n^2$.
2. For k to be an integer, n must divide 12 (the other terms are integers): $n \in \{1, 2, 3, 4, 6, 12\}$.
3. $n = 1$: $k = 12 + 9 - 1 = 20$. $n = 2$: $6 + 18 - 4 = 20$. $n = 3$: $4 + 27 - 9 = 22$. $n = 4$: $3 + 36 - 16 = 23$. $n = 6$: $2 + 54 - 36 = 20$. $n = 12$: $1 + 108 - 144 = -35$ (not positive).
4. Distinct positive values of k : 20, 22, 23.
5. Sum: $20 + 22 + 23 = 65$.

Answer: 65**37.**

SAT CHALLENGE · CALCULATOR

$$P = \frac{a}{\sqrt[5]{w^2}}$$

The formula above relates the positive quantities P and w , where a is a positive constant. Which of the following expresses w in terms of P and a ?

- (A) $w = \left(\frac{a}{P}\right)^{2/5}$ (B) $w = \left(\frac{P}{a}\right)^{2/5}$
 (C) $w = \left(\frac{a}{P}\right)^{5/2}$ ✓ (D) $w = \left(\frac{P}{a}\right)^{5/2}$

1. Write the root as a power: $\sqrt[5]{w^2} = w^{2/5}$, so $P = a w^{-2/5}$.
2. Multiply by $w^{2/5}$ and divide by P : $w^{2/5} = \frac{a}{P}$.
3. Raise both sides to the reciprocal power $\frac{5}{2}$: $w = \left(\frac{a}{P}\right)^{5/2}$.
4. (Numerical check: $a = 7$, $w = 3.2$ gives $P = 7 \cdot 3.2^{-0.4}$, and $\left(\frac{a}{P}\right)^{5/2} = 3.2$.)

Answer: (C)

38.For a constant k , the equation

$$2\sqrt{x+1} + 2\sqrt{15-x} = k$$

has exactly one real solution. What is the value of k ?

(A) 8

(B) $8\sqrt{2}$ ✓

(C) 16

(D) $2 + 2\sqrt{15}$

1. The domain is $-1 \leq x \leq 15$. The two expressions under the radicals add to 16 and are symmetric about $x = 7$ ($x + 1$ and $15 - x$ swap when $x \rightarrow 14 - x$), so if x is a solution then so is $14 - x$.
2. Exactly one solution forces $x = 14 - x$, i.e. $x = 7$.
3. $k = 2\sqrt{8} + 2\sqrt{8} = 4\sqrt{8} = 8\sqrt{2}$.
4. (Check: the left side rises from 8 at $x = -1$ to its maximum $8\sqrt{2} \approx 11.3$ at $x = 7$, then falls back to 8. Only the peak value is reached once. Other options give 0 or 2 solutions.)

Answer: (B)**39.**Positive real numbers x and y satisfy

$$\sqrt{x} - \sqrt{y} = 2 \quad \text{and} \quad x + y = 26.$$

What is the value of xy ?

1. Let $a = \sqrt{x}$ and $b = \sqrt{y}$. Then $a - b = 2$ and $a^2 + b^2 = 26$.
2. Square the first: $a^2 - 2ab + b^2 = 4$, so $26 - 2ab = 4$ and $ab = 11$.
3. $xy = a^2b^2 = (ab)^2 = 121$.
4. (Check: $a = 1 + 2\sqrt{3}$, $b = 2\sqrt{3} - 1$ gives $a - b = 2$, $a^2 + b^2 = 26$, $ab = 12 - 1 = 11$.)

Answer: 121

40.

SAT CHALLENGE · NO CALCULATOR NEEDED

For how many integer values of k with $-10 \leq k \leq 10$ does the equation

$$\sqrt{x+k} = x-2$$

have exactly one real solution?

1. Square both sides, remembering the condition $x-2 \geq 0$: $x+k = x^2 - 4x + 4$, so $x^2 - 5x + (4-k) = 0$ with roots $x = \frac{5 \pm \sqrt{9+4k}}{2}$.
2. Real roots need $9+4k \geq 0$; for integers this means $k \geq -2$. For $k \leq -3$ there are no solutions.
3. The larger root is at least $\frac{5}{2} > 2$, so it is always valid. The smaller root is valid only if $\frac{5 - \sqrt{9+4k}}{2} \geq 2$, i.e. $\sqrt{9+4k} \leq 1$, i.e. $k \leq -2$.
4. $k = -2$: the roots are 2 and 3, and both work, so there are two solutions. For $k \geq -1$ the smaller root is below 2 and is extraneous, leaving exactly one solution.
5. The values are $k = -1, 0, 1, \dots, 10$, which is 12 integers.

Answer: 12

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + bx^2 + cx + 12$ has integer coefficients b and c , and $x+3$ is a factor of $p(x)$. If p has exactly one real zero, what is the greatest possible value of c ?

1. $x+3$ is a factor, so $p(-3) = 0$: $-27 + 9b - 3c + 12 = 0 \Rightarrow c = 3b - 5$.
2. Divide: $p(x) = (x+3)(x^2 + ux + 4)$ with $u = b - 3$. (Expanding gives $x^3 + (u+3)x^2 + (3u+4)x + 12$, so $b = u+3$ and $c = 3u+4 = 3b-5$.)
3. Exactly one real zero means the quadratic factor has no real zero: $u^2 - 16 < 0$, a strict inequality, so $-4 < u < 4$.
4. $u = 4$ is excluded: $x^2 + 4x + 4 = (x+2)^2$ has the real zero -2 , giving two distinct real zeros of p (-3 and -2).
5. The greatest integer u is 3, so $b = 6$ and $c = 3(6) - 5 = 13$. (Including the boundary $u = 4$ would wrongly give $c = 16$.)

Answer: 13

42.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $6x^4 + kx^2 + 24$, where k is a constant, can be written as $(2x^2 + a)(3x^2 + b)$, where a and b are integers. What is the greatest possible value of k ?

1. Expand: $(2x^2 + a)(3x^2 + b) = 6x^4 + (2b + 3a)x^2 + ab$. So $ab = 24$ and $k = 3a + 2b$.
2. For a fixed product, $3a + 2b$ is largest when the factor with the bigger weight takes the bigger share, and one factor is as small as possible. Integer pairs (a, b) with $ab = 24$ include $(1, 24)$, $(24, 1)$, $(4, 6)$, $(6, 4)$, $(8, 3)$ and the negative pairs.
3. $(a, b) = (1, 24)$: $k = 3 + 48 = 51$. $(a, b) = (24, 1)$: $k = 72 + 2 = 74$. Other positive pairs: $(2, 12)$ gives 30, $(12, 2)$ gives 40, $(3, 8)$ gives 25, $(8, 3)$ gives 30, $(4, 6)$ gives 24, $(6, 4)$ gives 26.
4. Negative pairs make k negative. The greatest value is $k = 74$.

Answer: 74**43.**

SAT CHALLENGE · NO CALCULATOR NEEDED

For a real number a with $|a| \neq 1$,

$$\frac{1}{1-a} + \frac{1}{1+a} + \frac{2}{1+a^2} + \frac{4}{1+a^4} = 10.$$

What is the value of a^8 ?

- (A) $\frac{4}{5}$
(C) $\frac{9}{5}$

(B) $\frac{5}{4}$

(D) $\frac{1}{5}$ ✓

1. Pair the first two terms: $\frac{1}{1-a} + \frac{1}{1+a} = \frac{(1+a) + (1-a)}{1-a^2} = \frac{2}{1-a^2}$.
2. Add the third term: $\frac{2}{1-a^2} + \frac{2}{1+a^2} = \frac{2(1+a^2) + 2(1-a^2)}{1-a^4} = \frac{4}{1-a^4}$.
3. Add the fourth term: $\frac{4}{1-a^4} + \frac{4}{1+a^4} = \frac{4(1+a^4) + 4(1-a^4)}{1-a^8} = \frac{8}{1-a^8}$.
4. So $\frac{8}{1-a^8} = 10 \Rightarrow 1-a^8 = \frac{4}{5} \Rightarrow a^8 = \frac{1}{5}$.
5. ($\frac{4}{5}$ is the value of $1-a^8$, not of a^8 .)

Answer: (D)

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a real number k , consider the equation

$$x - 4\sqrt{x} = k.$$

For how many integer values of k from -10 to 10 , inclusive, does the equation have exactly one real solution x ?

1. Let $t = \sqrt{x}$ with $t \geq 0$. The equation becomes $t^2 - 4t - k = 0$; each root $t \geq 0$ gives exactly one $x = t^2$.
2. Discriminant: $16 + 4k$. Real roots need $k \geq -4$. Roots: $t = 2 \pm \sqrt{4 + k}$.
3. Case $k = -4$: double root $t = 2$, one solution ($x = 4$).
4. Case $k > 0$: product of roots $= -k < 0$, so one root is negative (rejected) and one is positive: exactly one solution.
5. Case $-4 < k \leq 0$: product $-k \geq 0$ and sum $4 > 0$, so both roots are nonnegative and distinct: two solutions. Case $k < -4$: none.
6. Valid integers: $k = -4$ and $k = 1, 2, \dots, 10$, which is $1 + 10 = 11$.

Answer: 11

45.

SAT CHALLENGE · NO CALCULATOR NEEDED

For real numbers a and b , $a^2 - b^2 = 36$. What is the value of $(3a + b)^2 - (a + 3b)^2$?

1. Do not expand. Treat it as $A^2 - B^2$ with $A = 3a + b$ and $B = a + 3b$.
2. $A - B = 2a - 2b = 2(a - b)$ and $A + B = 4a + 4b = 4(a + b)$.
3. Product: $8(a - b)(a + b) = 8(a^2 - b^2) = 8 \cdot 36 = 288$.
4. (Expanding confirms it: $9a^2 + 6ab + b^2 - a^2 - 6ab - 9b^2 = 8a^2 - 8b^2$.)

Answer: 288

46.

SAT CHALLENGE · NO CALCULATOR NEEDED

Let t be a positive real number with $t \neq 1$. Define $m = \sqrt[3]{t^2}$ and $n = \sqrt{t}$. If

$$\frac{m^p \cdot n^3}{\sqrt[6]{t}} = t^5,$$

what is the value of p ?

1. Rewrite as powers of t : $m = t^{2/3}$, $n = t^{1/2}$, $\sqrt[6]{t} = t^{1/6}$.

2. Left side: $\frac{t^{2p/3} \cdot t^{3/2}}{t^{1/6}} = t^{2p/3+3/2-1/6} = t^{2p/3+4/3}$.

3. Since $t > 0$ and $t \neq 1$, equal powers have equal exponents: $\frac{2p}{3} + \frac{4}{3} = 5 \Rightarrow \frac{2p}{3} = \frac{11}{3} \Rightarrow p = \frac{11}{2}$.

4. Enter 11/2 or 5.5.

Answer: 11/2

47.

SAT CHALLENGE · NO CALCULATOR NEEDED

The expression $\frac{(2x-3)^4(x+2)-7}{5}$ is equivalent to $ax^5 + bx^4 + cx^3 + dx^2 + ex + f$, where a, b, c, d, e and f are constants. What is the value of $a + b + c + d + e + f$?

(A) $-\frac{4}{5}$ ✓

(B) $\frac{3}{5}$

(C) -4

(D) 31

1. The sum of all the coefficients of a polynomial $p(x)$ equals $p(1)$, because every power of 1 is 1.

2. The expression is already equal to the polynomial, so evaluate it at $x = 1$: $(2 \cdot 1 - 3)^4 = (-1)^4 = 1$ and $(1 + 2) = 3$.

3. Numerator: $1 \cdot 3 - 7 = -4$. Dividing by 5: $\frac{-4}{5}$.

4. $a + b + c + d + e + f = -\frac{4}{5}$. (The other options are: $\frac{3}{5}$ forgets the -7 , -4 forgets the division by 5, and 31 is the constant term f , the value at $x = 0$.)

Answer: (A)

48.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^3 + (3 - k)x^2 - (10 + 3k)x + 10k = 0$$

In the equation above, k is a constant. The equation has exactly two distinct real solutions for exactly two values of k . What is the sum of these two values of k ?

1. Group the terms with k and the terms without: $(x^3 + 3x^2 - 10x) - k(x^2 + 3x - 10) = 0$.
2. Factor $x^3 + 3x^2 - 10x = x(x^2 + 3x - 10)$, so the equation is $(x^2 + 3x - 10)(x - k) = 0$.
3. $x^2 + 3x - 10 = (x + 5)(x - 2)$, so the equation is $(x + 5)(x - 2)(x - k) = 0$ with solutions -5 , 2 and k .
4. There are two distinct solutions only when k repeats one of the fixed roots: $k = 2$ or $k = -5$. (Otherwise there are three.)
5. Sum: $2 + (-5) = -3$.

Answer: -3